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Firewood, charcoal or electricity? Gender gaps in cooking, heating, and lighting energy use in Zambia

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ABSTRACT

While access to clean, affordable energy is crucial for well-being and sustainable development, energy poverty persists in developing countries. We examine the determinants and gender gaps in household energy use in Zambia across four key activities—cooking, heating, lighting, and boiling water—focusing on the most commonly used fuels: charcoal, electricity, firewood, and candles. Using survey data from 1779 households and nonlinear Oaxaca-Blinder decomposition methods, we find that education, income, and housing tenure are significant predictors of energy choices, with distinct gender patterns. Female-headed households—defined by a female decision-maker—are significantly more likely to use charcoal for cooking, heating, and boiling water, and to use candles for lighting. This pattern reflects a combination of socioeconomic disadvantages and structural constraints. By comparison, male-headed households predominantly use electricity, largely reflecting higher education and income among male decision-makers. Gender differences in firewood use are insignificant, though structural factors predominate. Robustness checks using Fairlie decomposition confirmed these gendered gaps stem from both resource endowments and behavioral norms. These findings underscore the need for gender-targeted policies that combine women's economic empowerment through education and income-generating opportunities, with infrastructure investments and affordability mechanisms to support equitable transition to clean energy in low-income contexts.

1. Introduction

Despite growing interest in the gender-energy nexus globally, much of the existing research in Africa has centered on the technical, economic, and environmental aspects of household energy transitions. Limited attention has been paid to the social and gendered dimensions (Adusah-Poku and Takeuchi, 2019; Kutortse, 2022; Ankrah et al., 2020; Hlahla, 2022). The few studies that engage with gender issues (e.g., Adusah-Poku and Takeuchi, 2019; Kutortse, 2022; Ankrah et al., 2020; Adusah-Poku et al., 2023; Cicowiez et al., 2022; Krishnapriya et al., 2021) tend to be concentrated in West or East African countries with higher electrification levels or different sociocultural dynamics. By contrast, Zambia's energy profile is characterized by significantly lower access to modern energy sources and a heavy reliance on biomass fuels, especially in rural areas. Compared to countries like Sri Lanka or China, Zambia lags considerably in energy infrastructure (e.g., power plants, transmission lines), while its gendered energy patterns may closely resemble those found in Tanzania or Ghana. These contextual distinctions highlight the need for localized gender-sensitive energy research tailored to Zambia's specific social, institutional, and infrastructural realities. While global studies provide useful reference points, their generalizability is constrained by differences in electrification rates, cultural norms, and policy environments. Our study addresses this gap by offering a rigorous, context-specific analysis of gender disparities in household energy use, with broader implications for developing countries where gender continues to shape energy-related decisions.

Understanding gender disparities in household energy use is not merely an academic exercise. It carries profound implications for sustainable development, public health, and gender equity. The Sustainable Development Goal 7 (SDG 7) calls for universal access to affordable,

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reliable, sustainable, and modern energy by 2030, while SDG 5 emphasizes gender equality and women's empowerment. These goals are inherently interconnected: energy poverty disproportionately affects women and girls who bear primary responsibility for household energy-related tasks, including cooking, heating, and fuel collection (International Energy Agency [IEA], 2023). Traditional energy systems that rely on biomass fuels impose significant health burdens through indoor air pollution—responsible for approximately 3.2 million premature deaths annually worldwide, with women and children most affected (World Health Organization [WHO], 2022). Moreover, time spent collecting firewood and managing inefficient cooking systems perpetuates "time poverty," constraining women's opportunities for education, income generation, and civic participation (Kyayesimira and Florence, 2021). Research shows that women's participation in household decision-making is associated with higher adoption rates of clean energy (Choudhuri and Desai, 2020; Rahut et al., 2014, 2017).

Addressing these gendered energy disparities requires a robust theoretical foundation. Drawing on feminist economics, energy justice frameworks, and theories of intrahousehold bargaining (Folbre, 1994; Agarwal, 1997), we examine how gendered power relations, structural inequalities, and decision-making autonomy shape household energy outcomes. While theoretical perspectives on gender and energy are diverse, ranging from feminist political ecology's emphasis on nature-society relations to energy justice's focus on distributional equity, they converge on a critical insight. Energy systems are not gender-neutral but rather reflect and reproduce existing power structures.

Therefore, this paper has three objectives. First, it identifies the determinants of household energy use across four essential domestic activities—cooking, heating, lighting, and boiling water focusing on the most prevalent fuels in households: charcoal, electricity, firewood, and candles. Second, we quantify disparities in these fuels for cooking, heating, lighting, and boiling water, based on the gender of a household's primary decision-maker, comparing male-headed and female-headed households. Third, we decompose these disparities into explained (observable endowments) and unexplained (structural or behavioral) components using nonlinear Oaxaca-Blinder decomposition methods (Yun, 2004) and the Fairlie decomposition technique (Fairlie, 2005) for robustness, thereby offering policy-relevant insights into the sources and drivers of gender gaps in energy use. The study draws on survey data from Zambia.

We classify households as male-headed (MHHs) or female-headed (FHHs) based on the gender (or sex) of the primary decision-maker, defined as the individual mainly responsible for decisions related to cooking, heating, and lighting energy use. This operationalization conceptually aligns with feminist economics frameworks that emphasize how decision-making authority determines resource allocation (Agarwal, 1997; Doss, 2013) and with energy justice perspectives that recognize decision-making power as central to energy access and outcomes (Clancy et al., 2020). While energy choices are made at the household level and households may include multiple male and female individuals, we focus on the gender of the primary decision-maker to explore how gendered authority shapes energy access and preferences. This classification reflects gendered roles in household energy decision-making, where the head's gender often shapes access to resources and fuel choices (Adusah-Poku et al., 2023; Ochieng et al., 2021).

We explicitly acknowledge important limitations of this binary categorization. First, it cannot capture intrahousehold bargaining dynamics, negotiation processes, or power gradients between household members—dimensions emphasized in feminist political ecology and bargaining literature (Agarwal, 1997; Doss, 2013). Second, it overlooks the roles and preferences of secondary decision-makers who may influence energy choices even without formal authority. Third, it treats 'male-headed' and 'female-headed' households as homogeneous categories, potentially obscuring within-group heterogeneity in decision-making arrangements. Despite these limitations, this approach provides a tractable and policy-relevant framework for analyzing systematic gender differences in energy outcomes at the household level. It produces results that are more interpretable for policy interventions, enables comparability with existing survey-based studies and aligns with how most national energy surveys operationalize gender, making our findings directly relevant for policy design (Zhang and Petrova, 2025). Future research employing household-level data or qualitative methods could provide the intrahousehold dynamics our approach was unable to capture. These conceptual and methodological considerations further underscore the importance of examining gendered energy dynamics in a context like Zambia, where energy poverty, low electrification, and entrenched gender roles make such disparities especially pronounced.

Zambia offers a compelling and underexplored setting for analyzing the gendered dynamics of household energy use. National electrification rates remain low, with only 46 % of the population having access to electricity, masking stark urban-rural disparities (85.7 % in urban areas versus 14.5 % in rural areas) (World Bank, 2020; Kannan et al., 2024). Moreover, 84 % of households rely on unclean fuels for cooking (World Bank, 2018). In this context, energy poverty is not gender neutral: traditional gender roles assign women and girls the responsibility for cooking and fuel collection, while limiting their involvement in household decision-making (Moniruzzaman and Day, 2020). Women and girls often travel long distances to collect firewood, reducing their opportunities for education and economic participation and exposing them to risks of violence and exploitation (Leduchowicz-Municio et al., 2023). As primary managers of household energy tasks, that is, cooking, heating, and lighting—women are disproportionately affected by dependence on traditional biomass fuels. This reliance not only increases exposure to indoor air pollution but also reinforces "time poverty," further constraining women's ability to pursue education, formal employment, and income-generating activities (Kyayesimira and Florence, 2021; Mangalsana Singh et al., 2024).

Grounded in feminist economics and energy justice theory, this study is timely and makes the following novel contributions to the literature on household energy use in developing countries. First, while the importance of gender of the decision maker in shaping household energy decisions is increasingly acknowledged in global energy discourse (e.g., Adusah-Poku and Takeuchi, 2019; Kutortse, 2022; Ankrah et al., 2020; Hlahla, 2022; Ngarava et al., 2022; Acheampong et al., 2021; Wen et al., 2023; Shen et al., 2025), what drives these disparities remain under-investigated in empirical work. Most prior studies either examine household fuel use without a gender lens or analyze gender-energy interactions without theoretically grounding their analysis in feminist economics or energy justice frameworks, nor do they decompose the roles of socioeconomic characteristics (endowments) versus structural barriers (discrimination, norms, and institutional biases). This limits the design of interventions that can effectively target the root sources of inequitable energy outcomes. This study, for the first time, comprehensively analyzes the determinants and gender gaps of fuel choice across four essential domestic activities—cooking, heating, lighting, and boiling across the gender lens—with particular attention to electricity and the fuel types prevalent in low-income settings (e.g., charcoal and firewood). We employ robust nonlinear decomposition methods, revealing structural inequalities in energy access and utilization for designing targeted and inclusive energy interventions.

Second, Zambia's energy landscape is characterized by low electrification and heavy reliance on traditional biomass fuels. National electricity access remains limited, with only 31.4 % of the population connected to the grid and rural electrification as low as 4.4 % (Ministry of Energy (MoE), 2019), forcing many households, both grid-connected and off-grid to depend on charcoal and firewood for cooking (MoE, 2019; Ng'ombe et al., 2025a,b). This dependence disproportionately burdens women and girls, who are primarily responsible for cooking and fuel collection—by increasing their exposure to indoor air pollution and

reducing the time available for education, employment, and income-generating activities (Chishimba and Muchapondwa, 2025). Candles and kerosene lamps, widely used for lighting, pose additional health and safety risks, particularly for women and children. Zambia's energy system is also highly vulnerable to climate variability, as evidenced by drought-induced reductions in Lake Kariba's reservoir levels, which have led to load shedding of up to 21 h daily (Zimba, 2024). In response to the energy crisis, Zambia's National Energy Policy (NEP) (MoE, 2019) prioritizes sustainable biomass utilization and gender mainstreaming. By analyzing the gendered dimensions of energy choices, this study offers novel insights into how energy poverty disproportionately affects women in low-income settings and provides lessons applicable to other energy-poor nations in Africa, Asia, and Latin America for advancing a gender-just energy transition in the Global South.

The remainder of this paper is organized as follows. Section 2 reviews relevant theoretical frameworks and empirical literature on the gender-energy nexus and positioning the study within current policy debates. Section 3 describes the data, variable construction, and descriptive statistics. Section 4 outlines the econometric methods used. Section 5 presents the empirical results and discusses policy implications. Section 6 concludes with key findings and recommendations for future research.

2. Literature review and theoretical framework

This section provides the theoretical foundation for our empirical analysis by synthesizing multiple complementary frameworks that explain different dimensions of gendered energy disparities. We draw particularly on feminist economics, household production theory, energy transition models, capabilities approach, and intersectionality theory. Together, these perspectives provide a robust lens for understanding why and how gender shapes household energy outcomes in contexts like Zambia.

Feminist economics reveals how gender norms and power asymmetries fundamentally shape resource allocation and energy choices. Gender roles, intra-household bargaining dynamics, and differential control over household finances determine consumption patterns in low-income contexts (Folbre, 1994; Agarwal, 1997). Women typically bear responsibility for cooking and fuel collection yet exercise limited control over household finances and major energy investments (Doss, 2013). Female-headed households, where women hold greater decision-making autonomy, may exhibit distinct energy preferences, though economic and institutional constraints frequently prevent adoption of modern fuels, even when preferred.

The household production model (Becker, 1965) treats energy as a productive input that can substitute for time, with gender-differentiated opportunity costs. This framework is particularly relevant where women disproportionately spend time on domestic tasks. In labor-market-active contexts, time savings from efficient modern energy represent significant opportunity costs, enabling participation in income generation and education. However, in rural labor-constrained settings where off-farm opportunities are limited, the primary welfare gains from efficient energy derive from reduced physical burden and improved health. They also expand time for care and education rather than income generation. This distinction matters for understanding whether gender energy gaps reflect access constraints, bargaining power deficits, or differences in returns to women's time in alternative activities.

Energy transition models explain how households move from traditional to modern fuels, though reality proves more complex than linear progression. The Energy Ladder Hypothesis (Leach, 1992) posits that households progress from traditional biomass to modern fuels as income and education rise. However, many households engage in "fuel stacking"—simultaneously using multiple fuels to balance affordability, cultural preferences, and supply reliability (Masera et al., 2000). Importantly, gender norms intersect with fuel transitions; women's reliance on traditional fuels often persists in better-off households, suggesting that income alone does not determine energy choices (Mulenga et al., 2019).

Capabilities approaches link energy access directly to human freedom and well-being, with particularly gendered implications. Drawing on Nussbaum (2011), this framework shows that energy poverty constrains capabilities by limiting education, health, and income opportunities. Because women and girls bear disproportionate burdens from fuel collection and indoor air pollution exposure, energy poverty reflects and reinforces gender inequality.

Intersectionality theory emphasizes that gender combines with other social categories to compound vulnerability. Gender intersects with income, education, and location to create multiple overlapping axes of marginalization (Crenshaw, 2013). Rural, less-educated women with insecure housing face particularly acute energy access constraints, as disadvantages compound rather than merely add.

Research across developing regions documents consistent gender-energy disparities, though with important contextual variations. In Asia, studies demonstrate that female-headed households experience higher multidimensional energy poverty due to limited labor-market participation, weaker social networks, and constrained asset ownership (Shen et al., 2025; Jayasinghe et al., 2025). Crucially, even where women prefer clean fuels, male-dominated financial decision-making restricts their influence over energy choices (Yasmin and Grundmann, 2020). Notably, Nepal et al. (2025) find that female education has stronger effects than male education in reducing urban energy poverty in China, underscoring women's empowerment as a critical lever for energy transitions.

While Asian contexts reveal these patterns, Sub-Saharan Africa exhibits similar gender-energy disparities with important regional variations shaped by infrastructure contexts and market structures. Studies across Tanzania, Ghana, and other African nations consistently demonstrate that female-headed households are more price-sensitive to electricity tariffs, more reliant on traditional biomass, and face binding economic constraints that prevent clean fuel adoption despite preferences (Wen et al., 2023; Adusah-Poku et al., 2023; Kutortse, 2022). Women's education and income emerge as significant factors increasing clean-fuel adoption probability across multiple contexts. Even in South Africa, with near-universal electrification, Ngarava et al. (2022) identify persistent gender gaps in energy access, suggesting infrastructure expansion alone is insufficient for energy equity.

A critical limitation across this literature is insufficient attention to whether gender energy disparities stem from observable characteristics—income, education, occupation—or unobservable structural factors including social norms, institutional biases, and intra-household power dynamics. Decomposition methods (e.g., Oaxaca-Blinder, Fairlie) address this gap by quantifying explained versus unexplained components of disparities. This distinction is essential for policy: endowment-based gaps call for resource transfers and capacity building, while structural gaps require addressing institutions and norms.

As mentioned before, within SDG frameworks, universal energy access (SDG 7) and gender equality (SDG 5) are interconnected imperatives. Zambia's National Energy Policy of 2019 embraces these priorities through commitments to sustainable biomass management, gender mainstreaming, and equitable modern energy access. Yet implementation gaps persist due to financing constraints, infrastructure deficits, and insufficient empirical evidence on gendered energy patterns. Emerging "just energy transition" frameworks emphasize that clean-energy policies must be socially inclusive and equitable (Clancy et al., 2020). Our analysis provides the rigorous empirical evidence necessary to inform gender-responsive energy strategies that simultaneously advance sustainability and equity with broader implications.

3. Data

3.1. Data sources

The data for this study were collected in the fourth quarter of 2020 through an in-person survey supported by the Increased Access to Electricity and Renewable Energy Production project and Zambia's Ministry of Energy. The survey was designed to provide social, economic, and environmental data to inform the implementation of the gender strategy under the revised 2019 National Energy Policy. The survey gathered information on household demographics, patterns of energy access, productive uses of electricity, and household perspectives, and preferences regarding the current energy landscape.

The sample comprised 1786 households drawn from six districts across five provinces—Chitambo and Serenje (Central), Ikelenge (Northwestern), Kalomo (Southern), Samfya (Luapula), and Lusaka

Table 1
Description of variables and descriptive statistics.

Variable Name	Variable Description	Total Sample	Std Dev	Female	Std Dev	Male	Std Dev	Mean Diff
Gender	Gender of the household's primary decision-maker (=1 if male, 0 o/w)	0.3901	0.4879					
Age	Age of respondent (years)	37.940	14.215	37.052	13.646	39.327	14.965	−2.274***
Marital status	Household head is married (=1, 0 o/w)	0.658	0.475	0.616	0.486	0.721	0.448	−0.105***
No education	Respondent does not have formal education (=1 if yes, 0 o/w)	0.060	0.237	0.072	0.259	0.040	0.196	0.032***
Primary education	Education level is from grade 1 to 7, (=1 0 o/w	0.352	0.478	0.403	0.490	0.270	0.444	0.132***
Secondary education	Education level is from grade 8 to 12, (=1, 0 o/w	0.349	0.477	0.346	0.476	0.353	0.478	−0.006
College education	Education: college certificate/diploma (=1, 0 o/w)	0.160	0.366	0.122	0.328	0.217	0.412	−0.094***
University education	Education: At least University degree (=1, 0 o/w	0.079	0.270	0.054	0.226	0.118	0.323	−0.063***
Income 500	Monthly income is less than ZMW500	0.318	0.466	0.338	0.473	0.286	0.452	0.051***
Income 500 to 2499	Monthly income is between ZMW500 to 2499	0.352	0.478	0.376	0.484	0.314	0.464	0.061***
Income 2500 to 4499	Monthly income is between ZMW2500 & 4499	0.137	0.344	0.138	0.345	0.134	0.340	0.004
Income over 4500	Monthly income at least ZMW4500	0.200	0.400	0.158	0.365	0.265	0.441	−0.106***
Household size	Number of household members	5.826	2.306	5.871	2.139	5.754	2.545	0.117
Farmer	Respondent's occupation is a farmer (=1 if yes, 0 o/w)	0.251	0.434	0.247	0.431	0.256	0.437	−0.009
Govt. employee	Respondent is a government employee (=1 if yes, 0 o/w)	0.154	0.361	0.125	0.331	0.198	0.399	−0.073***
Business employee	Respondent's occupation is business (=1 if yes, 0 o/w)	0.186	0.389	0.223	0.417	0.126	0.332	0.097***
Wage labor employee	Respondent's occupation is casual labor (=1 if yes, 0 o/w)	0.083	0.276	0.074	0.262	0.096	0.295	−0.021
Services employee	Occupation is cleaning, tutoring, hair dressing (=1, 0 o/w)	0.016	0.127	0.018	0.134	0.012	0.113	0.005
Retired employee	Respondent is retired (=1 if yes, 0 o/w)	0.033	0.178	0.023	0.150	0.047	0.212	−0.024***
House is rented	Respondent's house is rented (=1 if yes, 0 o/w)	0.247	0.432	0.263	0.440	0.221	0.415	0.041**
House is self-owned	The respondent's house is owned (=1 if yes, 0 o/w)	0.625	0.484	0.633	0.482	0.612	0.487	0.020
Employer owns house	Respondent's employer owns the house (=1 if yes, 0 o/w)	0.078	0.268	0.056	0.230	0.110	0.314	−0.054***
House owned other	Respondent's house owned in other means (=1 if yes, 0 o/w)	0.050	0.218	0.047	0.211	0.054	0.227	−0.007
Positive perception	Positive perception of electrification effects on community (=1 if yes, 0 o/w)	0.679	0.467	0.698	0.459	0.649	0.477	0.048***
Access to information	Access to renewable energy information on solar pumps (=1 if yes, 0 o/w)	0.297	0.457	0.237	0.425	0.390	0.488	−0.152***
Women's energy initiatives	Presence of community-based programs promoting clean energy adoption, such as cookstove distribution or solar access campaigns (=1 if yes, 0 o/w)	0.097	0.296	0.091	0.288	0.106	0.308	−0.015
Outcome variables								
Charcoal cooking	Household uses charcoal as fuel for cooking (=1 if yes, 0 o/w)	0.473	0.499	0.513	0.500	0.412	0.492	0.101***
Electricity for cooking	Household uses electricity as fuel for cooking (=1 if yes, 0 o/w)	0.146	0.353	0.117	0.321	0.191	0.393	−0.074***
Firewood for cooking	Household uses firwood as fuel for cooking (=1 if yes, 0 o/w)	0.379	0.485	0.369	0.482	0.396	0.489	−0.026
Charcoal for heating	Charcoal is the main energy source for heating (=1 if yes, 0 o/w)	0.478	0.499	0.511	0.500	0.426	0.494	0.085***
Electricity for heating	Electricity is the main energy source for heating (=1 if yes, 0 o/w)	0.117	0.321	0.100	0.301	0.143	0.350	−0.042***
Firewood for heating	Firewood is the main energy source for heating (=1 if yes, 0 o/w)	0.404	0.490	0.387	0.487	0.430	0.495	−0.043*
Candles for lighting	Candles are the main energy source for lighting (=1 if yes, 0 o/w)	0.169	0.375	0.192	0.394	0.134	0.340	0.058***
Electricity for lighting	Electricity is the main energy source for lighting (=1 if yes, 0 o/w)	0.516	0.499	0.517	0.499	0.514	0.500	0.002
Charcoal for boiling water	Charcoal is the main energy source for boiling water (=1 if yes, 0 o/w)	0.395	0.489	0.431	0.495	0.338	0.473	0.092***
Electricity for boiling water	Electricity is the main energy source for boiling water (=1 if yes, 0 o/w)	0.142	0.349	0.114	0.318	0.187	0.390	−0.073***
Firewood for boiling water	Firewood is the main energy source for boiling water (=1 if yes, 0 o/w)	0.462	0.498	0.454	0.498	0.474	0.498	−0.019
No. of observations		1779		1085		694		

Note: ***, **, *Statistically significant at 1 %, 5 % and 10 % significance levels, respectively.

(Lusaka Province), though in this study, we used a sample for 1779 households due to nonresponse observations. A multi-stage stratified sampling design ensured broad representation across urban and rural areas with varying levels of electrification.

Although the data were collected in 2020, they remain the most comprehensive and nationally representative dataset available on household energy access and use in Zambia. Since then, no subsequent national survey has collected comparable information linking energy choices with gender, income, education, and housing characteristics. Moreover, the structural features captured in the dataset—low electrification rates, heavy reliance on biomass, intermittent electricity supply, and persistent gender inequalities in energy access continue to define Zambia's current energy landscape (Musonda, 2025). The dataset therefore provides a robust and policy-relevant foundation for examining contemporary gendered energy patterns and informing ongoing clean energy transition efforts under the National Energy Policy.

3.2. Key variables and descriptive statistics

Table 1 summarizes the key variables used in our estimations, focusing on energy sources for cooking, heating, boiling water, and lighting in Zambia, alongside the explanatory variables. As mentioned earlier, gender refers to the sex (male or female) of the household's decision-maker regarding energy use. In Zambia, where patriarchal structures often centralize decision-making authority, the gender of the primary decision-maker captures critical differences in preferences, bargaining power and constraints shaping household energy behavior (Johnson et al., 2019). Thus, in a female-headed household, the decision-maker is female by sex. We find that 39 % of the decision-makers in our sample were male, consistent with surveys where women are more often available for interviews (Ochieng et al., 2021; Jeuland et al., 2015).

Demographic characteristics reveal gendered life-course patterns with energy implications (Table 1). Female heads are significantly younger and less likely to be married than their male counterparts. Age may shape energy preferences through generational exposure to modern technologies, while marital status affects household resource pooling. Married household heads often benefit from combined income and labor, facilitating investment in modern energy sources, whereas single or widowed women may face greater reliance on traditional fuels (Acheampong et al., 2021).

Educational attainment shows pronounced gender gaps, with female heads concentrated at primary levels while male heads dominate tertiary education. Education is a well-established determinant of energy choice, enhancing energy literacy, risk perception, and awareness of health and environmental risks associated with traditional biomass (Pachauri and Rao, 2013; Nepal et al., 2025). These educational disparities directly constrain women's energy decision-making capacity and economic opportunities.

Income patterns mirror educational gaps, with female heads over-represented in the lowest income brackets and male heads dominating higher income categories (Table 1). These differences align with the Energy Ladder Hypothesis (Leach, 1992), which predicts that income growth enables fuel transitions. However, persistent structural inequalities often limit such transitions for women, reinforcing "energy stacking" patterns in low-income contexts (Masera et al., 2000).

Household size, while not significantly different by gender of headship, captures total energy demand and economies of scale. Larger households may rely more on cheaper fuels to meet energy needs efficiently (Ankrah et al., 2020). In Zambian extended family contexts, household size also proxies for labor availability in firewood collection and cooking, tasks traditionally assigned to women and girls.

Occupational patterns function as proxies for both income stability and exposure to modern technologies. Table 1 shows that more male heads concentrated in government employment with predictable incomes and better electricity access, while female heads dominate self-employment and informal trading characterized by income volatility and liquidity constraints. Farming households of both genders show high firewood dependence, reflecting rural residence and limited grid access (Adusah-Poku and Takeuchi, 2019; Adusah-Poku et al., 2023; Ng'ombe et al., 2025).

Table 1 further shows that housing tenure is a crucial structural determinant. Female heads are more likely to rent, potentially limiting ability or incentive to invest in permanent energy infrastructure, while male heads benefit disproportionately from employer-provided housing typically equipped with electricity connections. Housing ownership enhances both security and long-term investment capacity (Cicowiez et al., 2022; Martey, 2019). Moreover, perceptual and informational variables illuminate behavioral dimensions often overlooked in energy studies. Despite female heads expressing stronger positive perceptions of electrification, they report significantly lower access to renewable energy information, particularly regarding solar technologies (Table 1). This paradox where women value modern energy more but access information less, reflects gendered information asymmetries observed across African contexts (Krishnapriya et al., 2021). It also suggests that awareness gaps reinforce existing access inequities.

Community-level indicators, such as women's energy initiatives promoting clean cookstoves or solar lanterns, provide institutional context for energy transitions. These programs, often supported by non-governmental organizations or development agencies, enhance women's participation in energy decision-making and facilitate cleaner fuel adoption (Hlahla, 2022). Their inclusion helps distinguish between economic and structural barriers in our decomposition analysis.

Taken together, these variables align closely with the theoretical foundations discussed earlier. Gender, age, education, income, occupation, and housing reflect economic endowments within the household production and energy ladder frameworks, while perception, information access, and women's initiatives capture behavioral and institutional dimensions emphasized in feminist and capabilities approaches. Integrating these diverse covariates enables a subtle understanding of how demographic, economic, and social factors jointly determine household energy behavior and contribute to observed gender disparities in Zambia.

4. Modeling and estimation techniques

4.1. Econometric framework

Following Hirvonen (2016), Addai et al. (2022), Chaudhuri et al. (2020), Avenyo et al. (2021), Asiseh et al. (2024) and Ng'ombe et al. (2025), we apply an extension of the Oaxaca-Blinder decomposition (Blinder, 1973; Oaxaca, 1973) proposed by Yun (2004). The intuition behind the nonlinear Oaxaca–Blinder decomposition is that it allows us to break down the gender gap in energy use into two meaningful parts, even when the outcome is modeled with nonlinear methods (e.g., logit or probit). The idea is simple: imagine two identical households except that one is male-headed and the other is female-headed. Part of the difference in their predicted energy use comes from differences in observable characteristics such as education, income, or dwelling quality. This is the "explained" portion. The rest comes from differences in how these characteristics are rewarded or converted into outcomes across gender groups. For example, unequal access to information, differences in electricity reliability, or social norms may restrict women's energy choices. This is the "unexplained" portion. The nonlinear extension simply ensures that this logic holds even when the relationship between the predictors and the outcome is not linear.

Following Powers et al. (2011), the standard problem of decomposing difference in the first moments in which the dependent variable is function of linear predictors and regression coefficients is

$$Y = G(X\beta) \tag{1}$$

where Y is an $n \times 1$ vector of dependent variables, β is a $k\times$ 1 vector of coefficients, X is an $n \times k$ matrix of covariates presented in Table 1, and $G(.)$ a differentiable function that maps a linear combination of $X\beta$ on Y where Y represents our outcome variables (i.e., household uses charcoal as fuel for cooking, household uses electricity as fuel for cooking, household uses firewood as fuel for cooking, see others in Table 1). The average difference in Y being male decision-makers ($\mathscr{M}$) and female decision-makers ($\mathscr{F}$) can be defined as:

$$\overline{Y}_{\mathscr{F}} - \overline{Y}_{\mathscr{M}} = \overline{G(X_{\mathscr{F}}\beta_{\mathscr{F}})} - \overline{G(X_{\mathscr{M}}\beta_{\mathscr{M}})} = \underbrace{\{\overline{G(X_{\mathscr{F}}\beta_{\mathscr{F}})} - \overline{G(X_{\mathscr{M}}\beta_{\mathscr{F}})}\}}_{Endowments\ (E)} + \underbrace{\{\overline{G(X_{\mathscr{M}}\beta_{\mathscr{F}})} - \overline{G(X_{\mathscr{M}}\beta_{\mathscr{M}})}\}}_{Coefficients\ (C)} \quad (2)$$

where *Endowments* (E) also referred to as the explained component, captures the portion of the gender gap in energy use outcomes that can be attributed to differences in observable characteristics between male- and female-headed households. These characteristics may include education, income, household size, access to infrastructure, and other measurable factors that influence energy use decisions. In contrast, the *Coefficients* (C) component—also known as the unexplained component reflects the mean increase Y for female household heads if they had the same characteristics as male household heads. Rather than stemming from observable differences, this component captures structural influences rather than observable differences. These may include discrimination, institutional barriers, social norms, or other unobserved constraints that limit female-headed households from benefiting equally from similar endowments (Aweke and Navrud, 2022).

In the decomposition presented in Equation (2), we use female decision-makers—FHHs (group $\mathscr{F}$) as the comparison group and male decision-makers—MHHs (group $\mathscr{M}$) as the reference group because they represent the conventional benchmark in gender-gap analyses. Using female-headed households as the reference would not change the substantive interpretation—only the sign of the estimated gaps. This allows us to interpret the unexplained component as the portion of the gender gap that would remain even if female heads had the same characteristics as male-heads, highlighting potential structural inequalities in energy use. Following Powers et al. (2011), the E component can be interpreted as a counterfactual comparison from the perspective of male-headed households.

Conversely, the C component reflects a counterfactual from the perspective of female heads—it estimates the expected difference if they exhibited the same behavioral response to covariates as male heads. This interpretation is symmetric; reversing the reference and comparison groups yields an equivalent decomposition, as shown in earlier work by Neumark (2004) and Oaxaca and Ransom (1988). As noted by Powers et al. (2011), the choice of the mapping function $G(.)$ in the decomposition depends on the nature of the dependent variable. Since our outcome variables—cooking, heating, and lighting energy choices—are binary, we adopt the logit model for the decomposition analysis.

While the aggregate decomposition provides insights into the overall explained and unexplained components of the gender gap, it is also important to understand which specific covariates drive these differences. To this end, we perform a detailed decomposition, where the E and C components are further disaggregated into E_j and C_j, respectively. Each jth component corresponds to the individual contribution of the jth covariate (for $j = 1, \ldots, J$) to the explained and unexplained portions of the gap. Moreover, it might seem intuitive to compute each component E_j and C_j by directly substituting one group's covariates (or coefficients) with those of the other group. However, in nonlinear decompositions, such as those based on logit models, the results are sensitive to the order in which covariates are introduced—a problem known as path dependence (Yun, 2004). This contrasts with linear decomposition methods, where the order of substitution does not affect the results. To address this issue, Fairlie (2005) proposed a method known as sequential covariate replacement with randomization of the ordering of covariate substitutions. While effective, this approach can be computationally intensive and is primarily suited for binary outcome models with a modest number of covariates. Further details of the nonlinear OB decomposition are provided in the Appendix.

4.2. Robustness check

As a robustness check, we complement the nonlinear Oaxaca-Blinder decomposition with the Fairlie decomposition method. The Fairlie decomposition follows the same logic as the nonlinear Oaxaca-Blinder decomposition but is specifically designed for binary outcomes, such as whether a household uses electricity or relies on firewood. Instead of comparing regression coefficients, it compares predicted probabilities from separate models for male- and female-headed households. The intuition is to ask: if female-headed households had the same characteristics as male-headed households, would their probability of using a cleaner fuel increase, and by how much? By sequentially substituting characteristics and tracking how predicted probabilities change, the Fairlie method quantifies how much of the gender gap in a binary outcome is due to differences in characteristics versus differences in how those characteristics shape choices. This makes it especially useful in energy-access settings and is well-suited for binary outcomes, allowing for a sequential replacement of covariates to address path dependence. By comparing results across both methods, we assess the consistency of our findings regarding gender disparities in household energy use. Additional details of the Fairlie decomposition method are in the Appendix.

5. Results and discussion

In this section, we first present estimation results of the determinants of energy fuel choices for cooking, heating, boiling water and lighting by gender of a household's primary decision-maker in Zambia. Nonlinear decomposition results are also presented.

5.1. Determinants of adoption of energy fuels for cooking

Table 2 shows logit model results of gendered patterns in the use of charcoal, electricity, and firewood for cooking. Among male heads, primary education is associated with lower charcoal use relative to no education, while higher income (ZMW500 and above) shows positive associations, consistent with Mulenga et al. (2019). Occupation correlates strongly: farmers are less likely to use charcoal, possibly due to cost constraints, while businessmen and retirees show higher likelihood. Renting is positively associated with charcoal use, while access to renewable energy information correlates with reduced use (Rahut et al., 2017).

Female household heads show somewhat different patterns. Older age and secondary education correlate positively with charcoal use, as do monthly incomes up to ZMW4,499. Female farmers and government employees are less likely to use charcoal. Notably, positive perceptions of electricity among female heads are significantly associated with continued charcoal reliance a counterintuitive finding that warrants closer examination. This pattern illustrates what aspiration-constraint theory describes as the gap between preferences and realized capabilities (Ray, 2006). Women may value electricity's benefits (cleanliness, convenience, safety) yet face binding affordability constraints that prevent adoption (Adjei-Mantey et al., 2022). Within our feminist economics framework, this reflects how women's limited control over household finances and constrained access to credit prevent them from translating preferences into energy choices, even when they hold decision-making authority. Zambia's unreliable electricity supply due to load-shedding (Zimba, 2024; Bwalya Umar & Kunda-Wamuwi, 2019; Bayliss & Pollen, 2021) may make charcoal a rational second-best choice even among those who prefer electricity. This finding underscores that

Table 2
Logit models for energy sources for cooking by gender.

Variable/Model	Charcoal Model		Electricity Model		Firewood Model	
	Male	Female	Male	Female	Male	Female
	Coefficient	Coefficient	Coefficient	Coefficient	Coefficient	Coefficient
Explanatory variables						
Age	0.001	0.012*	0.017	−0.005	−0.011	−0.016**
	(0.007)	(0.006)	(0.011)	(0.011)	(0.009)	(0.008)
Marital status	−0.273	−0.132	−0.955***	−0.249	1.622***	0.394**
	(0.215)	(0.152)	(0.281)	(0.235)	(0.320)	(0.199)
Primary education	−0.948**	0.344	13.567	−1.024	0.679	−0.294
	(0.456)	(0.291)	(807.137)	(0.658)	(0.515)	(0.328)
Secondary education	−0.368	1.132***	14.586	0.400	−0.103	−1.790***
	(0.449)	(0.309)	(807.137)	(0.584)	(0.511)	(0.366)
College education	−0.586	0.540	15.164	1.192*	−0.784	−3.318***
	(0.499)	(0.391)	(807.137)	(0.626)	(0.632)	(0.685)
University education	−0.608	−0.494	15.403	2.189***	−2.325***	−2.519***
	(0.531)	(0.437)	(807.137)	(0.649)	(0.873)	(0.742)
Income 500 to 2499	1.172***	0.786***	−0.753*	−0.471	−1.138***	−0.640***
	(0.244)	(0.178)	(0.401)	(0.367)	(0.299)	(0.212)
Income 2500 to 4499	0.850***	0.618***	0.298	0.714**	−1.378***	−1.341***
	(0.309)	(0.238)	(0.407)	(0.363)	(0.408)	(0.326)
Income over 4500	0.621**	0.414	0.889**	0.912**	−2.357***	−1.758***
	(0.292)	(0.257)	(0.366)	(0.369)	(0.437)	(0.403)
Household size	−0.017	0.012	−0.125**	−0.142**	0.094*	0.061
	(0.037)	(0.035)	(0.063)	(0.070)	(0.048)	(0.042)
Farmer	−0.947***	−2.145***	−1.813***	−1.293**	1.250***	2.503***
	(0.284)	(0.237)	(0.657)	(0.651)	(0.324)	(0.269)
Govt. employee	0.353	−0.643**	−0.137	0.614*	−0.128	1.023*
	(0.335)	(0.296)	(0.394)	(0.363)	(0.560)	(0.525)
Business	0.809***	−0.092	−0.322	0.009	−1.045***	0.217
	(0.284)	(0.196)	(0.378)	(0.339)	(0.393)	(0.241)
Wage labor	−0.127	0.217	0.558	0.323	−0.254	−0.422
	(0.308)	(0.304)	(0.425)	(0.417)	(0.410)	(0.448)
Services	0.313	0.988	−0.111	−0.384	−0.830	−1.282
	(0.725)	(0.782)	(0.879)	(1.088)	(1.021)	(1.127)
Retired	1.304***	0.000	0.138	1.203*	0.000	−1.674
	(0.489)	(0.515)	(0.598)	(0.640)	(0.000)	(1.102)
House is rented	0.724*	0.475	1.071**	1.431**	−3.291***	−3.178***
	(0.395)	(0.353)	(0.517)	(0.561)	(0.602)	(0.548)
House is self-owned	0.129	−0.525	0.494	0.751	−1.003**	−0.189
	(0.382)	(0.338)	(0.513)	(0.562)	(0.488)	(0.414)
Employer owns house	0.656	0.565	−0.190	−1.225*	−0.454	0.781
	(0.449)	(0.447)	(0.584)	(0.729)	(0.608)	(0.547)
Explanatory variables						
Positive perception	0.114	0.297*	0.056	−0.358	−0.150	−0.159
	(0.186)	(0.164)	(0.250)	(0.256)	(0.260)	(0.215)
Access to information	−0.422**	−0.253	0.461*	0.381	0.222	0.140
	(0.181)	(0.180)	(0.239)	(0.266)	(0.247)	(0.241)
Women's energy initiatives	−0.002	−0.417	−1.920***	−0.019	1.159***	0.700**
	(0.291)	(0.269)	(0.649)	(0.458)	(0.366)	(0.333)
Constant	−0.474	−0.773	−16.171	−2.391**	0.490	1.003
	(0.658)	(0.535)	(807.137)	(0.941)	(0.799)	(0.656)
Loglikelihood values	−403.348	−572.829	−236.605	−256.750	−240.207	−374.841
Observations	694	1085	694	1085	661	1085

Note: Standard errors in parentheses. No formal education is used as a reference group. Income less than ZMW500 is used as a reference group. ***, **, *Statistically significant at 1 %, 5 % and 10 % significance levels, respectively.

favorable attitudes alone are insufficient for clean energy transitions without addressing economic barriers and infrastructure reliability. It also suggests that awareness campaigns, while necessary, must be complemented by affordability interventions such as subsidized appliances, payment plans, or targeted income support for FHHs.

For electricity use, we find that electricity use for cooking shows strong associations with education and monthly income, particularly among female heads. Among male heads, marriage, farming, and larger household sizes are associated with lower electricity use, while higher monthly income (above ZMW4,500), renting housing, and access to information are positively associated. For female heads, college or university education and monthly incomes above ZMW2,500 are positively associated with electricity use, while farming and larger household size are associated with lower use. Government employment, retirement, and renting are associated with greater electricity use among female heads.

Firewood use reflects notable gender patterns. Among male heads, marriage and larger households correlate with greater firewood reliance, while higher education and all income levels are associated with reduced use. Male farmers show greater firewood reliance. Among female heads, older age is associated with lower firewood use, while marriage correlates with higher use. Education and income show consistent negative associations with firewood use (Mulenga et al., 2017). Female heads who are farmers and government employees are more likely to use firewood, a pattern that likely reflects occupational deployment to semi-rural or peri-urban areas where electricity access remains limited, rather than purely residential location (Bwalya Umar et al., 2022). Renters, by contrast, are less likely to use firewood, possibly due to restrictions on fuel use imposed by landlords.

Cooking energy choices reflect intersections of gender, education,

income, and occupation. Consistent with the Energy Ladder Hypothesis, higher education and income promote shifts toward electricity, but male-headed households benefit more due to greater economic and educational advantages. Female-headed households often face an aspiration-constraint gap: positive attitudes toward electricity coexist with reliance on charcoal, highlighting constrained agency shaped by limited financial control and barriers to credit. The household production model explains why larger households and farming occupations rely on traditional fuels, considering differing opportunity costs of time and labor. From a capabilities perspective, restricted access to clean energy limits women's ability to pursue health, education, and income opportunities. Effective interventions must address both resource gaps (education, income) and structural barriers (credit access, infrastructure reliability, information asymmetries).

5.2. Determinants of adoption of energy fuels for heating

Table 3 presents logit model results of gendered associations in the use of charcoal, electricity, and firewood for heating, revealing patterns broadly consistent with cooking but with some notable differences. Monthly income and occupation emerge as key correlates. Among male heads, higher monthly income (ZMW500–ZMW2,499) is positively associated with charcoal use, while farming correlates negatively. Businessmen and retirees show higher likelihood of charcoal use, while access to renewable energy information correlates with lower use. Female heads show similar patterns: primary and secondary education correlate positively with charcoal use, as does monthly income.

Electricity use for heating shows gender-specific associations. Among male heads, older age correlates positively with electricity use,

Table 3
Logit models for energy sources for heating by gender.

Variable/Model	Charcoal Model		Electricity Model			
	Male	Female	Male	Female	Firewood Model	Female
	Coefficient	Coefficient	Coefficient	Coefficient	Firewood Model	Coefficient
Explanatory variables						
Age	−0.002	0.009	0.031***	−0.003	−0.017**	−0.013**
	(0.007)	(0.006)	(0.012)	(0.011)	(0.007)	(0.006)
Marital status	0.290	−0.154	−1.168***	−0.076	0.369	0.194
	(0.216)	(0.144)	(0.308)	(0.242)	(0.243)	(0.162)
Primary education	−0.606	0.491*	12.668	−0.824	0.224	−0.397
	(0.455)	(0.282)	(471.208)	(0.642)	(0.453)	(0.290)
Secondary education	0.102	0.794***	12.880	0.536	−0.450	−1.103***
	(0.449)	(0.297)	(471.208)	(0.584)	(0.448)	(0.311)
College education	−0.076	0.626	14.147	1.312**	−1.448***	−2.155***
	(0.500)	(0.381)	(471.208)	(0.635)	(0.530)	(0.491)
University education	−0.757	−0.343	14.636	2.333***	−1.165**	−2.029***
	(0.535)	(0.427)	(471.208)	(0.654)	(0.573)	(0.548)
Income 500 to 2499	1.184***	0.893***	−0.541	−0.486	−1.079***	−0.838***
	(0.235)	(0.168)	(0.440)	(0.369)	(0.242)	(0.178)
Income 2500 to 4499	0.846***	0.538**	0.448	0.329	−1.127***	−0.813***
	(0.303)	(0.224)	(0.442)	(0.385)	(0.332)	(0.257)
Income over 4500	0.780***	0.260	0.483	0.564	−1.201***	−0.694**
	(0.288)	(0.244)	(0.400)	(0.392)	(0.317)	(0.291)
Household size	−0.054	−0.011	−0.053	−0.055	0.089**	0.035
	(0.038)	(0.033)	(0.063)	(0.065)	(0.040)	(0.035)
Farmer	−0.580**	−1.292***	−2.141***	−1.091*	1.082***	1.461***
	(0.269)	(0.210)	(0.787)	(0.589)	(0.265)	(0.211)
Govt. employee	0.231	−0.154	−0.152	0.710*	0.183	0.031
	(0.336)	(0.288)	(0.413)	(0.383)	(0.401)	(0.383)
Business	0.758***	0.127	−0.268	0.180	−0.659**	−0.110
	(0.286)	(0.190)	(0.408)	(0.349)	(0.319)	(0.205)
Wage labor	0.152	0.247	−0.150	0.709*	−0.015	−0.673*
	(0.308)	(0.287)	(0.496)	(0.416)	(0.337)	(0.364)
Services	0.189	0.726	−0.282	0.000	0.011	−0.255
	(0.730)	(0.657)	(0.929)	(0.000)	(0.842)	(0.685)
Retired	1.078**	0.242	−0.562	1.280*	0.000	0.000
	(0.481)	(0.508)	(0.632)	(0.658)	(0.000)	(0.000)
House is rented	0.060	0.176	1.197**	3.082***	−0.973**	−1.586***
	(0.388)	(0.340)	(0.609)	(1.080)	(0.449)	(0.395)
House is self-owned	−0.557	−0.749**	1.051*	2.920***	−0.032	−0.142
	(0.374)	(0.325)	(0.607)	(1.078)	(0.412)	(0.353)
Employer owns house	0.203	−0.127	−0.402	0.583	0.166	0.677
	(0.445)	(0.422)	(0.714)	(1.226)	(0.500)	(0.459)
Explanatory variables						
Positive perception	0.053	0.271*	0.011	−0.481*	−0.012	−0.051
	(0.182)	(0.154)	(0.272)	(0.257)	(0.204)	(0.172)
Access to information	−0.305*	−0.329*	0.402	−0.005	0.142	0.393**
	(0.179)	(0.169)	(0.264)	(0.282)	(0.198)	(0.191)
Women's energy initiatives	−0.074	−0.101	−1.915**	−0.263	0.701**	0.252
	(0.287)	(0.251)	(0.776)	(0.513)	(0.302)	(0.271)
Constant	−0.325	−0.503	−16.244	−4.956***	0.771	1.022*
	(0.654)	(0.510)	(471.209)	(1.299)	(0.684)	(0.550)
Loglikelihood value	−412.458	−630.270	−201.911	−247.762	−352.895	−526.161
Observations	692	1081	692	1061	692	1081

Note: Standard errors in parentheses. No formal education is used as a reference group. Income less than ZMW500 is used as a reference group. ***, **, *Statistically significant at 1 %, 5 % and 10 % significance levels, respectively.

while marriage and farming show negative associations. Renting or owning housing correlates positively, possibly reflecting better infrastructure access in urban areas. College and university education show strong positive associations, while farming correlates negatively, consistent with lower rural electrification rates (Kannan et al., 2024; Ng'ombe et al., 2025).

Female heads who are government employees, retirees, and renters or homeowners show higher likelihood of electricity use. Unexpectedly, female heads with positive perceptions of electricity are less likely to use it for heating—an association that challenges simple attitude-behavior models. Several factors may explain this pattern: (1) end-use prioritization: households with positive electricity attitudes but limited access may reserve scarce electricity for higher-priority uses (lighting, phone charging) rather than heating, which can be met with cheaper biomass alternatives (Munro and Schiffer, 2019); (2) appliance costs: electric heaters represent significant capital investment that may be prohibitive even for grid-connected households; (3) climate context: Zambia's tropical climate reduces heating needs relative to cooking and lighting, making heating a lower priority for electricity adoption even among those who value electricity generally. This finding highlights that energy preferences are differentiated by end use and that positive attitudes toward a fuel do not automatically translate into adoption across all household activities.

Firewood use for heating remains common, particularly in rural areas (Mazzone et al., 2021), though education and income show consistent negative associations. Among male heads, older age and higher education correlate with reduced firewood use, while larger households and farming show positive associations. Increased monthly income is consistently associated with reduced firewood use across categories. Businessmen and renters are less likely to rely on firewood. Female heads show similar associations: higher education and income correlate with lower firewood use, while farmers remain heavy users. Interestingly, access to information on renewable energy (such as solar pumps) also positively correlates with firewood use. This suggests that rural outreach efforts may require refinement to ensure they promote transitions beyond biomass efficiency improvements.

Heating energy results highlight education and income as key drivers of clean energy transitions, while exposing complex attitude-behavior gaps. Among female heads, positive views of electricity often fail to translate into use, as budget constraints and competing household priorities limit adoption—challenging the household production model's rational choice assumption. Feminist economics explains this constrained agency: women's decision-making authority is undermined by limited purchasing power. Persistent firewood reliance among farmers and rural households reflects spatial inequalities in infrastructure, aligning with capabilities theory's notion of restricted opportunity sets. Zambia's tropical climate further shapes energy hierarchies, with heating ranked below cooking and lighting in resource allocation. Effective, gender-responsive heating policies must address end-use differentiation, appliance affordability, and rural-urban infrastructure disparities to close gaps between preferences and actual fuel use.

5.3. Determinants of adoption of energy fuels for lighting

To conserve space, logit model results for determinants of energy fuels for lighting and boiling water are presented in Table A1 in the Appendix. Based on Table A1, patterns of candle usage reveal important insights into energy deprivation and socioeconomic disparities among households. Among male heads, candle use correlates positively with retirement, moderate monthly income, and rented accommodation—factors likely reflecting economic or infrastructural limitations. Retired individuals may lack consistent electricity access or capital for electrical lighting investments, while renters may face landlord restrictions on electrical connections. Conversely, university-level education and information access show negative associations with candle use. This suggests that awareness and knowledge facilitate transitions toward safer, modern alternatives (Chishimba and Muchapondwa, 2025).

Among female heads, older age and moderate income correlate positively with candle use, possibly reflecting age-related vulnerabilities and limited disposable income. However, female farmers and those with greater information access are less likely to use candles, indicating that occupation and informational empowerment can influence energy choices.

Electricity for lighting serves as a strong indicator of socioeconomic advancement. Among male heads, electricity use is positively associated with older age, higher education, and greater income relationships consistent with both financial means and knowledge required to access utility services (Mulenga et al., 2019). However, marriage, farming, public employment, and employer-provided housing show negative associations, potentially capturing constrained agency or rigid housing arrangements limiting energy decisions. Female heads show even stronger associations between electricity use and socioeconomic variables. Education at all levels—from primary to university—correlates substantially with increased electricity use, highlighting education's potential role in addressing energy poverty. Income growth and secure housing arrangements (renting or owning) also support electricity access. Farming and women's energy initiatives again show negative associations, reinforcing patterns observed across other energy uses.

In sum, lighting energy results show the strongest links between socioeconomic factors and energy outcomes, underscoring lighting's central role in household welfare. A pronounced education gradient—especially among female heads—supports human capital and capabilities theories, with women's education proving particularly powerful for electricity adoption across all levels (Nepal et al., 2025). Feminist economics highlights that lighting decisions, requiring less capital than cooking appliances, allow women's education to directly improve household outcomes despite limited financial control. Yet older female heads and renters remain vulnerable to candle dependence, illustrating intersectionality's insight into compounded disadvantages from gender, age, and housing insecurity. Information access emerges as a key enabler, suggesting that reducing gendered information asymmetries could accelerate lighting transitions even under income constraints.

5.4. Determinants of adoption of energy fuels for boiling water

Boiling water represents humanity's universal daily energy requirement, serving essential functions from food preparation to hygiene. Table A1 presents results for gendered determinants of energy use for boiling water, which largely mirror patterns observed for cooking and heating.

Charcoal shows widespread use, particularly among households with intermediate income levels. Among male heads, higher monthly income and retirement correlate with greater charcoal use, consistent with moderate-income reliance on transitional fuels (Mulenga et al., 2019). Business ownership also shows positive associations. Conversely, higher education (primary, college, and university levels), farming, and renewable energy information access correlate with reduced charcoal use.

Female heads show similar patterns: moderate income correlates with greater charcoal use, while secondary education shows positive and university education negative associations. Renting correlates positively with charcoal use, likely reflecting urban affordability constraints. Marriage, farming, information access, and women's energy initiatives show negative associations with charcoal use.

For electricity, among male heads, older age, renting, home ownership, and information access correlate positively with electricity use. Lower income and farming show negative associations, highlighting persistent rural and economic barriers (Adjei-Mantey et al., 2022). Female heads show positive associations between electricity use and secondary-to-university education, higher income, retirement, and

stable housing, consistent with Mulenga et al. (2019). Farming remains negatively associated with electricity use across gender categories.

Firewood use remains common among resource-constrained households. Among male heads, marriage, primary education, farming, and larger household size correlate with greater firewood use. Higher education, income, non-agricultural employment, and renting show negative associations, indicating transitions away from traditional fuels where feasible.

Female heads show similar associations: marriage, farming, and public sector employment correlate positively with firewood use, while higher education, income, wage labor, and rental housing show negative associations. Older age among women correlates with reduced firewood use, possibly reflecting preferences for more convenient energy sources. Overall, firewood use is consistently associated with lower education and income levels and traditional livelihoods (Mulenga et al., 2019). While female heads show positive attitudes toward cleaner energy, access remains constrained by education and income (Ochieng et al., 2021).

Boiling water energy patterns mirror cooking, highlighting education and income as key drivers of fuel choice. Consistency across cooking, heating, and boiling supports the Energy Ladder Hypothesis but reveals limits: income growth alone cannot ensure clean transitions without infrastructure, credit, and information. Moderate-income reliance on charcoal reflects "fuel stacking" (Masera et al., 2000), where households balance cost, convenience, and availability rather than moving linearly up the ladder. Boiling's short, frequent use makes it highly sensitive to cost and convenience trade-offs. Female-headed households' continued dependence on traditional fuels despite education and positive electricity attitudes underscores feminist economics' view of structural constraints—limited financial resources and access hinder transitions. Overall, Zambia's household energy shifts remain shaped by intertwined gender, income, and educational inequalities, requiring multifaceted policy responses.

Taken together, although our findings do not establish causality, the consistent patterns across cooking, heating, lighting, and boiling water reveal that Zambia's household energy transitions remain shaped by intertwined gender, income, and educational inequalities. Education and income emerge as the strongest enablers of clean energy adoption, underscoring the importance of improving women's education, financial inclusion, and energy literacy. Continued reliance on charcoal and firewood among FHHs reflects structural barriers—limited electrification access, low and unstable incomes, and credit constraints rather than preferences. A puzzling pattern is worth mentioning: women's energy initiatives correlate negatively with male-headed electricity use and positively with female-headed firewood reliance. Possible explanations include program targeting toward female-led households, emphasis on biomass efficiency (e.g., cookstove distribution) rather than full transitions, measurement limitations in cross-sectional data, and substitution effects where male-headed households maintain electricity while female-headed households optimize constrained biomass use. These associations likely reflect program design and targeting rather than discouragement of electricity adoption.

5.5. Nonlinear decomposition results

This section presents nonlinear Oaxaca-Blinder decomposition results that partition observed gender gaps in energy use into components attributable to differences in observable characteristics (explained) versus differences in behavioral patterns or structural factors (unexplained). We emphasize that these decompositions identify statistical associations rather than causal mechanisms, and that unexplained components may reflect behavioral differences, unobserved constraints, or structural barriers.

5.5.1. Cooking

Table 4 presents decomposition results for cooking fuels between households whose primary decision-maker is male and those whose primary decision-maker is female. Significant gender disparities emerge, with contributions from both observable characteristics (explained effects) and discrimination (unexplained effects).

For charcoal, the total gap (−0.1013) is statistically significant at 1 % significance level, indicating that MHHs are 10.13 percentage points less likely to use charcoal than FHHs. Of this gap, 23.41 % is attributable to differences in observable characteristics, while 76.59 % remains unexplained. The dominant unexplained component (76.59 %) suggests that behavioral differences, preferences, or systemic barriers disproportionately affecting FHHs drive charcoal reliance beyond measurable socioeconomic factors. Among explained factors, household income contributes substantially, with the ZMW500–ZMW2,499 bracket accounting for 15.82 % of the explained gap, consistent with income-related fuel transitions (Mulenga et al., 2019). Education shows mixed contributions: primary education narrows the gap as better-educated male decision-makers show greater charcoal use, while secondary education widens it.

For electricity, the pro-male gap of 7.46 percentage points, statistically significant at 1 % significance level, is primarily attributable to differences in characteristics (75.90 %), while the unexplained component (C = 0.0180) is statistically insignificant. Higher education—particularly college and university attainment—accounts for much of the explained gap, as male heads more frequently attain these levels. This pattern suggests that reducing educational disparities could narrow the electricity access gap. Primary education among female heads contributes negatively to the gap, indicating that even basic education is associated with higher electricity use among FHHs. Marital status and income differences also contribute to the explained component.

For firewood, the gender gap is small and statistically insignificant (0.0267), with the unexplained component dominating (110.96 %). Marital status contributes significantly to the unexplained portion, potentially reflecting differences in household responsibilities or traditional practices. Education shows complex associations: secondary and college education among male heads contribute to unexplained increases in firewood use, possibly linked to rural residence patterns. Household size also appears relevant, with larger MHHs showing greater firewood reliance through unexplained channels.

5.5.2. Heating

Table 5 presents decomposition results for heating fuels. For charcoal, the total gap is negative and statistically significant (−0.0845), with an overwhelming 96.67 % attributable to unexplained factors—the highest proportion observed across all fuel-use combinations. This suggests that behavioral, cultural, or structural factors rather than observable characteristics dominate charcoal heating disparities. While marital status and primary education show large, explained effects, these are statistically insignificant. The significant unexplained components for marital status, farming, and business ownership indicate that even among observably similar groups, male decision-makers are less likely to use charcoal for heating.

Electricity for heating shows a pro-male gap of 4.20 percentage points—statistically significant at 1 % significance level, is largely attributable to characteristic differences (89.45 %). Education is the dominant factor: college and university education contribute positively to the explained gap, suggesting that equalizing educational attainment would substantially narrow the disparity. These associations align with prior findings linking education to energy access and technological adoption (Ochieng et al., 2021). Marital status and age also contribute: younger male heads widen the gap, while marriage narrows it. Notably, women's energy initiatives in communities are associated with narrowing the electricity gap through the explained component.

For firewood, the statistically significant (at 5 % significance level) pro-male gap (0.0422) exhibits an unusual pattern: the unexplained component exceeds 100 % (161.43 %), while the explained component

Table 4
Nonlinear Oaxaca-Blinder decomposition of the male-female gap in energy fuels for cooking.

Variable/Model	Charcoal				Electricity				Firewood			
	Percent		Percent		Percent		Percent		Percent		Percent	
Panel 1: Gender gap and its sources												
Explained: due to diff. in characteristics	−0.0237**	23.41			0.0566***	75.90			−0.0029	−10.96		
	(0.0119)				(0.0079)				(0.0105)			
Unexplained: due to diff. in coefficients	−0.0776***	76.59			0.0180	24			0.0296	110.96		
	(0.0239)				(0.0152)				(0.0200)			
Total	−0.1013***				0.0746***				0.0267			
	(0.0212)				(0.0155)				(0.0163)			
Panel 2: Detailed decomposition												
	Explained		Unexplained		Explained		Unexplained		Explained		Unexplained	
Age	0.0004	−0.35	−0.0852	84.18	0.0028	3.70	−0.0260	−34.80	−0.0013	−4.82	−0.0133	−49.82
	(0.0037)		(0.0761)		(0.0018)		(0.0377)		(0.0039)		(0.0340)	
Marital status	−0.0064	6.27	−0.0183	18.11	−0.0072***	−9.62	0.0137	18.40	0.0040	14.88	0.0595**	223.21
	(0.0047)		(0.0353)		(0.0022)		(0.0195)		(0.0115)		(0.0303)	
Primary education	0.0278**	−27.44	−0.1102**	108.84	−0.1262***	−169.15	−0.1827	−244.86	−0.0016	−5.94	0.0247	92.61
	(0.0130)		(0.0444)		(0.0104)		(0.2372)		(0.0051)		(0.0217)	
Secondary education	−0.0005	0.52	−0.1099***	108.51	0.0066***	8.89	−0.1524	−204.25	−0.0001	−0.20	0.0395*	148.08
	(0.0006)		(0.0398)		(0.0005)		(0.1954)		(0.0002)		(0.0214)	
College education	−0.0123	12.14	−0.0292*	28.80	0.1011***	135.50	−0.0531	−71.14	−0.0027	−10.07	0.0212**	79.66
	(0.0101)		(0.0166)		(0.0081)		(0.0678)		(0.0085)		(0.0094)	
University education	−0.0086	8.46	−0.0013	1.29	0.0689***	92.43	−0.0222	−29.82	−0.0042	−15.60	−0.0006	−2.07
	(0.0072)		(0.0079)		(0.0057)		(0.0285)		(0.0127)		(0.0046)	
Income 500 to 2499	−0.0160***	15.82	0.0306	−30.23	0.0033**	4.46	0.0034	4.49	0.0016	5.96	−0.0125	−46.96
	(0.0033)		(0.0227)		(0.0016)		(0.0066)		(0.0048)		(0.0124)	
Income 2500 to 4499	−0.0008***	0.79	0.0068	−6.71	−0.0001	−0.12	0.0018	2.43	0.0001	0.51	0.0003	1.06
	(0.0003)		(0.0119)		(0.0001)		(0.0031)		(0.0004)		(0.0062)	
Income over 4500	0.0146**	−14.44	0.0069	−6.86	0.0068**	9.06	0.0001	0.15	−0.0058	−21.69	−0.0067	−25.31
	(0.0069)		(0.0133)		(0.0030)		(0.0025)		(0.0175)		(0.0091)	
Household size	0.0004	−0.42	−0.0359	35.44	0.0010**	1.40	−0.0031	−4.10	−0.0003	−1.12	0.0217	81.25
	(0.0009)		(0.0606)		(0.0005)		(0.0170)		(0.0009)		(0.0318)	
Farmer	−0.0020***	1.96	0.0626***	−61.78	−0.0012***	−1.64	0.0040	5.43	0.0004	1.36	−0.0191*	−71.53
	(0.0007)		(0.0239)		(0.0004)		(0.0079)		(0.0011)		(0.0112)	
Govt. employee	0.0057	−5.66	0.0264**	−26.06	−0.0007	−0.96	0.0030	3.98	0.0005	1.97	−0.0084	−31.38
	(0.0054)		(0.0126)		(0.0021)		(0.0042)		(0.0019)		(0.0071)	
Business	−0.0174***	17.15	0.0427**	−42.16	0.0022	2.99	0.0023	3.13	0.0017	6.44	−0.0186*	−69.78
	(0.0060)		(0.0171)		(0.0027)		(0.0043)		(0.0051)		(0.0102)	
Wage labor	−0.0006	0.60	−0.0054	5.35	0.0009	1.17	−0.0006	−0.74	−0.0000	−0.08	0.0018	6.66
	(0.0015)		(0.0069)		(0.0006)		(0.0015)		(0.0002)		(0.0035)	
Services	−0.0004	0.37	−0.0026	2.60	0.0000	0.06	−0.0002	−0.21	0.0001	0.28	0.0010	3.62
	(0.0009)		(0.0042)		(0.0005)		(0.0011)		(0.0003)		(0.0020)	
Retired	0.0071***	−6.97	0.0063*	−6.27	0.0002	0.32	0.0008	1.04	0.0000	0.00	0.00000	0.00
	(0.0025)		(0.0036)		(0.0010)		(0.0012)		(0.000)		(0.0000)	
House is rented	−0.0067*	6.58	0.0139	−13.73	−0.0032*	−4.27	0.0030	4.01	0.0033	12.30	−0.0026	−9.69
	(0.0040)		(0.0318)		(0.0018)		(0.0081)		(0.0097)		(0.0173)	
House is self-owned	−0.0006	0.58	0.0875	−86.41	−0.0007	−0.98	0.0051	6.87	0.0005	2.00	−0.0438	−164.40
	(0.0019)		(0.0731)		(0.0009)		(0.0184)		(0.0016)		(0.0372)	
Employer owns house	0.0079	−7.83	0.0011	−1.06	−0.0007	−0.99	−0.0018	−2.46	−0.0007	−2.49	−0.0058	−21.91
	(0.0058)		(0.0081)		(0.0024)		(0.0029)		(0.0021)		(0.0042)	
Positive perception	−0.0012	1.22	−0.0270	26.67	−0.0002	−0.26	−0.0091	−12.24	0.0001	0.40	0.0014	5.12
	(0.0020)		(0.0365)		(0.0009)		(0.0146)		(0.0004)		(0.0181)	
Access to information	−0.0142**	14.06	−0.0085	8.41	0.0050*	6.73	−0.0006	−0.80	0.0009	3.28	0.0026	9.80
	(0.0062)		(0.0129)		(0.0028)		(0.0027)		(0.0025)		(0.0064)	
Women's energy initiatives	−0.0000	0.01	0.0080	−7.90	−0.0021***	−2.82	0.0055	7.33	0.0004	1.67	0.0035	13.04
	(0.0010)		(0.0075)		(0.0007)		(0.0071)		(0.0013)		(0.0037)	
Constant	0.0000		0.0632	−62.44	0.0000		0.4269	572.27	0.0000		−0.0161	−60.31
	(0.0000)		(0.1805)		(0.0000)		(0.5464)		(0.0000)		(0.0786)	
Observations	1779		1779	1779	1779		1779	1779	1779		1779	

Notes: Standard errors in parentheses. No formal education is used as a reference group. Income less than ZMW500 is used as a reference group. ***, **, *Statistically significant at 1 %, 5 % and 10 % significance levels, respectively. The percent column gives the contribution of each variable in the model to the overall difference in respective energy fuel's use between male and female headed households.

Table 5
Nonlinear Oaxaca-Blinder decomposition of the male-female gap in energy fuels for heating.

Variable/Model	Charcoal				Electricity				Firewood			
		Percent		Percent		Percent		Percent		Percent		Percent
Panel 1: Gender gap and its sources												
Explained: due to diff.	−0.0028	3.33			0.0376***	89.45			−0.0259**	−61.43		
in characteristics	(0.0120)	96.67			(0.0083)				(0.0117)			
Unexplained: due to	−0.0817***				0.0044	10.55			0.0681***	161.43		
diff in coefficients	(0.0245)				(0.0145)				(0.0233)			
Total	−0.0845***				0.0420***				0.0422**			
	(0.0219)				(0.0135)				(0.0198)			
Panel 2: Detailed decomposition												
	Explained		Unexplained		Explained		Unexplained		Explained		Unexplained	
Age	0.0118	−13.92	−0.0769	91.03	0.0041***	9.70	−0.0072	−17.21	−0.0072**	−17.14	−0.0258	−61.16
	(0.7408)		(0.0690)		(0.0013)		(0.0257)		(0.0030)		(0.0517)	
Marital status	−0.0860	101.76	0.0545*	−64.56	−0.0072***	−17.09	0.0038	9.11	0.0071	16.91	0.0159	37.81
	(5.3579)		(0.0320)		(0.0017)		(0.0139)		(0.0048)		(0.0267)	
Primary education	−0.2264	267.99	−0.0881**	104.29	−0.1020***	−242.87	−0.0320	−76.31	−0.0055	−12.94	0.0371	88.01
	(14.1893)		(0.0422)		(0.0109)		(0.1138)		(0.0111)		(0.0320)	
Secondary education	−0.0019	2.20	−0.0477	56.47	0.0051***	12.04	−0.0253	−60.19	−0.0005	−1.27	0.0335	79.49
	(0.1142)		(0.0370)		(0.0006)		(0.0894)		(0.0005)		(0.0276)	
College education	0.0204	−24.15	−0.0171	20.27	0.0811***	193.03	−0.0093	−22.05	−0.0252***	−59.87	0.0128	30.41
	(1.3413)		(0.0157)		(0.0101)		(0.0327)		(0.0089)		(0.0126)	
University education	0.1357	−160.62	−0.0045	5.30	0.0563***	134.03	−0.0039	−9.40	−0.0136**	−32.34	0.0069	16.48
	(8.5344)		(0.0074)		(0.0070)		(0.0140)		(0.0065)		(0.0062)	
Income 500 to 2499	0.2061	−243.93	0.0218	−25.76	0.0019	4.57	0.0000	0.05	0.0123***	29.07	−0.0134	−31.81
	(12.9032)		(0.0208)		(0.0015)		(0.0011)		(0.0031)		(0.0168)	
Income 2500 to 4499	0.0101	−11.95	0.0085	−10.03	−0.0001	−0.27	−0.0001	−0.25	0.0009***	2.08	−0.0064	−15.26
	(0.6321)		(0.0105)		(0.0001)		(0.0006)		(0.0003)		(0.0086)	
Income over 4500	−0.2339	276.90	0.0164	−19.42	0.0030	7.05	0.0001	0.18	−0.0235***	−55.71	−0.0119	−28.25
	(14.6420)		(0.0118)		(0.0025)		(0.0005)		(0.0063)		(0.0103)	
Household size	−0.0177	20.97	−0.0503	59.56	0.0004	0.84	−0.0001	−0.32	−0.0019**	−4.53	0.0464	109.97
	(1.1085)		(0.0548)		(0.0004)		(0.0029)		(0.0009)		(0.0469)	
Farmer	0.0155	−18.31	0.0350*	−41.40	−0.0012**	−2.79	0.0015	3.59	0.0019***	4.46	−0.0138	−32.83
	(0.9686)		(0.0180)		(0.0005)		(0.0055)		(0.0004)		(0.0129)	
Govt. employee	−0.0477	56.42	0.0096	−11.36	−0.0005	−1.26	0.0006	1.48	0.0025	5.87	0.0028	6.70
	(3.0039)		(0.0113)		(0.0018)		(0.0022)		(0.0054)		(0.0103)	
Business	0.2070	−245.04	0.0281*	−33.26	0.0014	3.30	0.0006	1.45	0.0118**	27.87	−0.0182	−43.22
	(12.9910)		(0.0155)		(0.0022)		(0.0022)		(0.0057)		(0.0127)	
Wage labor	−0.0093	11.06	−0.0014	1.67	−0.0002	−0.39	0.0004	0.90	−0.0001	−0.15	0.0073	17.25
	(0.5842)		(0.0063)		(0.0007)		(0.0013)		(0.0014)		(0.0054)	
Services	0.0029	−3.44	−0.0020	2.33	0.0000	0.00	0.0000	0.00	−0.0000	−0.03	0.0007	1.73
	(0.1830)		(0.0034)		(0.0000)		(0.0000)		(0.0008)		(0.0030)	
Retired	−0.0743	87.95	0.0038	−4.54	−0.0008	−1.83	0.0002	0.58	0.0000	0.00	0.0000	0.00
	(4.6551)		(0.0032)		(0.0008)		(0.0009)		(0.0000)		(0.0000)	
House is rented	0.0071	−8.39	−0.0060	7.15	−0.0029*	−6.88	0.0028	6.66	0.0074**	17.65	0.0239	56.71
	(0.4506)		(0.0275)		(0.0016)		(0.0101)		(0.0034)		(0.0227)	
House is self-owned	−0.0325	38.52	0.0243	−28.73	−0.0013	−3.02	0.0067	16.04	0.0001	0.29	0.0104	24.56
	(2.0372)		(0.0634)		(0.0008)		(0.0244)		(0.0016)		(0.0508)	
Employer owns house	−0.0312	36.92	0.0037	−4.37	−0.0013	−3.06	0.0003	0.74	0.0017	3.94	−0.0043	−10.09
	(1.9541)		(0.0072)		(0.0024)		(0.0012)		(0.0050)		(0.0056)	
Positive perception	0.0073	−8.66	−0.0303	35.82	−0.0000	−0.04	−0.0019	−4.54	0.0001	0.26	0.0040	9.45
	(0.4652)		(0.0328)		(0.0008)		(0.0071)		(0.0018)		(0.0276)	
Access to information	0.1311	−155.15	0.0011	−1.34	0.0035	8.44	−0.0006	−1.32	0.0040	9.44	−0.0088	−20.92
	(8.2303)		(0.0118)		(0.0024)		(0.0019)		(0.0055)		(0.0097)	
Women's energy initiatives	0.0032	−3.80	0.0005	−0.57	−0.0017**	−4.08	0.0009	2.03	0.0020**	4.69	0.0061	14.38
	(0.2017)		(0.0073)		(0.0008)		(0.0030)		(0.0008)		(0.0055)	
Constant			0.0354	−41.86			0.0669	159.32			−0.0371	−87.99
			(0.1650)				(0.2362)				(0.1290)	
Observations	1779		1779	1779	1779		1779	1779	1779		1779	1779

Notes: Standard errors in parentheses. No formal education is used as a reference group. Income less than ZMW500 is used as a reference group. ***, **, *Statistically significant at 1 %, 5 % and 10 % significance levels, respectively. The percent column gives the contribution of each variable in the model to the overall difference in respective energy fuel's use between male and female headed households.

is negative (−61.43 %), indicating that characteristic differences favor FHHs but are more than offset by unexplained factors. Higher education among male heads (college and university) contributes to reduced firewood reliance through the explained component, consistent with educational transitions away from traditional fuels (Chishimba and Muchapondwa, 2025). Income effects are mixed: while higher income is associated with reduced firewood use (Bahadur et al., 2017), middle-income brackets (ZMW500–ZMW2,499) show greater reliance, possibly reflecting cost-conscious behavior. Farming increases the explained gap, likely due to rural concentration, while renting is associated with higher firewood use through both explained and unexplained channels, possibly because renters less frequently own energy-efficient appliances (Souza, 2018).

5.5.3. Lighting

Table 6 presents decomposition results for lighting fuels. For candles, male heads are 5.86 percentage points less likely to rely on candles than female heads—statistically significant at 1 % significance level, with

Table 6
Nonlinear Oaxaca-Blinder decomposition of the male-female gap in energy fuels for lighting.

Variable/Model	Candles				Electricity			
VARIABLES	Percent		Percent		Percent		Percent	
Panel 1: Gender gap and its sources								
Explained: due to diff. in characteristics	−0.0285***	48.65			0.0076	−285.99		
	(0.0110)				(0.0122)			
Unexplained: due to diff. in coefficients	−0.0301	51.35			−0.0102	385.99		
	(0.0214)				(0.0245)			
Total	−0.0586***				−0.0026			
	(0.0168)				(0.0204)			
Panel 2: Detailed decomposition								
	Explained		Unexplained		Explained		Unexplained	
Age	0.0054**	−9.17	0.0236	−40.19	0.0064	−240.88	0.0917	−3473.16
	(0.0026)		(0.0385)		(0.0048)		(0.1746)	
Marital status	−0.0033	5.61	−0.0083	14.21	−0.0081	306.40	−0.1104	4179.03
	(0.0037)		(0.0208)		(0.0054)		(0.1758)	
Primary education	0.0029	−4.95	−0.0166	28.37	0.0034	−128.87	−0.0861	3257.70
	(0.0096)		(0.0287)		(0.0076)		(0.1371)	
Secondary education	−0.0002	0.42	−0.0023	3.88	0.0004	−14.75	−0.0941	3561.00
	(0.0005)		(0.0240)		(0.0005)		(0.1487)	
College education	0.0013	−2.20	0.0057	−9.69	0.0169	−640.28	−0.0054	205.33
	(0.0071)		(0.0102)		(0.0139)		(0.0249)	
University education	−0.0072	12.31	−0.0000	0.01	0.0111	−421.48	−0.0099	376.45
	(0.0057)		(0.0053)		(0.0093)		(0.0181)	
Income 500 to 2499	−0.0056**	9.49	0.0161	−27.38	−0.0025	94.42	−0.0069	263.05
	(0.0026)		(0.0137)		(0.0024)		(0.0341)	
Income 2500 to 4499	−0.0003	0.52	0.0087	−14.92	−0.0004	16.82	0.0072	−273.32
	(0.0002)		(0.0074)		(0.0003)		(0.0196)	
Income over 4500	−0.0006	0.99	0.0043	−7.34	0.0116	−439.60	−0.0041	156.71
	(0.0052)		(0.0089)		(0.0084)		(0.0192)	
Household size	0.0003	−0.48	0.0209	−35.59	−0.0007	25.20	0.0219	−829.59
	(0.0006)		(0.0363)		(0.0007)		(0.0941)	
Farmer	−0.0008*	1.29	−0.0020	3.48	−0.0015	55.04	0.0223	−843.30
	(0.0004)		(0.0108)		(0.0010)		(0.0467)	
Govt. employee	0.0009	−1.57	0.0074	−12.62	−0.0099	374.79	−0.0285	1077.08
	(0.0042)		(0.0086)		(0.0074)		(0.0457)	
Business	0.0006	−1.00	−0.0068	11.62	−0.0048	182.87	0.0220	−831.78
	(0.0044)		(0.0102)		(0.0040)		(0.0432)	
Wage labor	−0.0006	0.95	0.0004	−0.72	−0.0007	24.97	−0.0083	314.66
	(0.0011)		(0.0040)		(0.0009)		(0.0156)	
Services	−0.0005	0.88	0.0007	−1.26	−0.0003	9.65	0.0044	−165.43
	(0.0005)		(0.0017)		(0.0005)		(0.0081)	
	Explained		Unexplained		Explained		Unexplained	
Retired	0.0000	0.00	0.0001	−67.06	0.0006	−21.56	−0.0042	158.04
	(0.0000)		(0.0001)		(0.0016)		(0.0079)	
House is rented	−0.0091**	15.55	0.0393*	−151.72	−0.0030	113.77	−0.0584	2210.71
	(0.0041)		(0.0238)		(0.0028)		(0.0906)	
House is self-owned	−0.0023	3.88	0.0889	−12.21	0.0002	−7.71	−0.1342	5080.53
	(0.0019)		(0.0601)		(0.0011)		(0.2168)	
Employer owns house	0.0026	−4.44	0.0072	59.33	−0.0083	315.38	−0.0079	300.01
	(0.0064)		(0.0070)		(0.0064)		(0.0160)	
Positive perception	0.0021	−3.59	−0.0348	16.71	−0.0041	156.08	0.0297	−1122.83
	(0.0015)		(0.0234)		(0.0027)		(0.0698)	
Access to information	−0.0148***	25.33	−0.0098	−0.25	0.0026	−98.87	0.0111	−421.33
	(0.0047)		(0.0082)		(0.0045)		(0.0233)	
Women's energy initiatives	0.0007	−1.18	0.0001	294.68	−0.0014	52.62	−0.0011	42.13
	(0.0007)		(0.0043)		(0.0010)		(0.0106)	
Constant	0.0000		−0.1727		0.0000		0.3391	−12835.69
	(0.0000)		(0.1258)		(0.0000)		(0.5777)	
Observations	1779		1779	1779	1779		1779	1779

Notes: Standard errors in parentheses. No formal education is used as a reference group. Income less than ZMW500 is used as a reference group. ***, **, **Statistically significant at 1 %, 5 % and 10 % significance levels, respectively. The percent column gives the contribution of each variable in the model to the overall difference in respective energy fuel's use between male and female headed households.

observable characteristics accounting for 48.65 % of the gap and unexplained factors for 51.35 % (though the latter is not statistically significant). The substantially explained component suggests that characteristic differences are important contributors.

Access to renewable energy information is the most influential explained factor, accounting for 25.33 % of the total gap, indicating that male heads' greater information access is associated with significantly lower candle use. Housing tenure also contributes: male heads are more likely to rent, which is associated with reduced candle reliance through the explained component (15.55 %), though the positive unexplained component for renting suggests unobserved factors partially offset this. Income in the ZMW500–ZMW2,499 bracket contributes negatively to the explained gap, as better economic standing is associated with reduced candle use. Age contributes positively, while farming engagement contributes marginally to reducing the gap through the explained component.

For electricity lighting, the gender gap is negligible and statistically insignificant (−0.0026), with neither explained (0.0076) nor unexplained (−0.0102) components reaching significance. The large relative percentages result from the very small absolute gap and should be interpreted cautiously. Overall, electricity access for lighting appears relatively equitable between MHHs and FHHs.

5.5.4. Boiling water

Table 7 presents decomposition results for boiling water. For charcoal, the statistically significant gap (−0.0927) at 1 % significance level indicates male heads are 9.27 percentage points less likely to use charcoal, with characteristic differences accounting for 43 % and unexplained factors for 57 %. Education contributes substantially: primary education shows positive explained effects but strong negative unexplained effects, suggesting that even with similar education levels, male heads are less likely to use charcoal. Higher education (college and university) is associated with reduced charcoal use through the explained component. Income effects are mixed: moderate income (ZMW500–ZMW2,499) contributes to reduced charcoal use among male heads through the explained component, though unexplained factors partially offset this, while higher income shows positive explained effects reflecting substitution patterns observed by Mulenga et al. (2019). Information access and business occupation contribute through the explained component. Housing tenure shows large unexplained effects, suggesting gendered behaviors linked to housing stability (Souza, 2018).

For electricity, the pro-male gap of 7.30 percentage points is largely attributable to characteristic differences (76.32 %). This gap is statistically significant at 1 % significance level. Education is the strongest contributor: college (E = 0.0970) and university education (E = 0.0669) significantly contribute to the explained gap, reinforcing findings by Mulenga et al. (2019) and Chishimba and Muchapondwa (2025). Primary education narrows the gap, suggesting FHHs with basic education may prioritize electricity for safety or convenience. Marital status also narrows the gap through the explained component. Information access and stable housing tenure contribute to narrowing the gap, suggesting these factors support FHHs' transitions to electricity. The relatively small unexplained component indicates limited behavioral differences once access constraints are accounted for.

For firewood, the gender gap is small and statistically insignificant (0.0197), with 98.61 % attributable to unexplained factors. Despite the lack of statistical significance, education—particularly secondary and college levels—shows associations with increased firewood use among MHHs through unexplained channels. Marital status and housing tenure also contribute via unexplained effects, suggesting that traditional norms, family structure, and logistical factors influence fuel choices beyond observable economic factors, as observed in Brazil by Mazzone et al. (2021).

5.5.5. Synthesis of decomposition findings and policy implications

The decomposition analyses reveal three overarching patterns in gendered energy disparities. First, traditional fuel gaps (charcoal and firewood) are predominantly driven by unexplained factors, with 76.59 % (charcoal for cooking), 96.67 % (charcoal for heating), and 56.64 % (charcoal for boiling) of gaps attributable to behavioral differences or structural barriers rather than observable characteristics. This suggests that female-headed households' greater reliance on traditional fuels reflects factors beyond education, income, or occupation—potentially including social norms, risk preferences, credit constraints, or unmeasured infrastructure barriers. These large unexplained components indicate that resource transfers alone will not eliminate gender energy disparities; institutional reforms addressing norms, credit access, and discriminatory practices are essential. Second, electricity gaps are primarily explained by observable characteristics: 75.90 % (cooking), 89.45 % (heating), and 76.32 % (boiling water) of electricity gaps stem from characteristic differences, with education and income emerging as the dominant factors. This pattern suggests that equalizing educational attainment and income between male- and female-headed households could substantially reduce gender disparities in electricity access. The relatively small unexplained components indicate limited behavioral differences once access constraints are accounted for, pointing toward human capital investments as the most direct pathway to gender equity in electricity access.

Third, mixed and unexpected patterns emerge in middle-income effects and education-firewood associations, where moderate-income households sometimes show greater traditional fuel use, and educated male heads exhibit unexplained firewood reliance—patterns that may reflect rural residence, fuel stacking strategies, or measurement limitations. These findings suggest that policies promoting clean energy transitions must address both resource gaps (education, income, infrastructure) and structural barriers (norms, credit access, information asymmetries) to achieve gender-equitable energy access.

5.6. Robustness checks

To ensure robustness of the nonlinear Oaxaca-Blinder decomposition results, we conducted Fairlie decomposition as an alternative method for analyzing male-female gaps in energy fuel choices. This approach, specifically designed for nonlinear models, decomposes differences in binary outcomes between groups; results appear in Table A2 (Appendix). The Fairlie decomposition broadly confirms patterns identified in the main analysis, reinforcing the presence of persistent gender gaps and the role of key socioeconomic factors.

The Fairlie results show statistically significant gender differences across all energy uses, closely mirroring main findings. In cooking, FHHs are 10.12 percentage points more likely to use charcoal and 7.45 percentage points less likely to use electricity than MHHs, with a smaller marginally significant gap in firewood (0.0266). For heating, FHHs show greater charcoal reliance (gap = −0.0852) and lower electricity use (gap = 0.0422), with a statistically significant firewood gap favoring MHHs (0.0430, $p < 0.05$). For lighting, FHHs rely more on candles (gap = −0.0586, $p < 0.01$), while electricity gaps are negligible and insignificant. For boiling water, FHHs are 9.27 percentage points more likely to use charcoal and 7.30 percentage points less likely to use electricity, with no significant firewood difference (0.0196).

These consistent trends affirm the robustness of our findings: FHHs are more reliant on traditional fuels while MHHs more frequently use electricity. The Fairlie decomposition also consistently identifies socioeconomic characteristics—particularly education, income, housing status, and information access—as accounting for significant portions of observed gender disparities in household energy use.

6. Conclusion and policy implications

This study investigates the determinants of household energy use across four essential domestic activities—cooking, heating, lighting, and boiling water in Zambia. We focus on the most common fuels in low-

Table 7
Nonlinear Oaxaca-Blinder decomposition of the male-female gap in energy fuels for boiling water.

Variable/Model	Charcoal				Electricity				Firewood			
		Percent		Percent		Percent		Percent		Percent		Percent
Panel 1: Gender gap and its sources												
Explained: due to diff. in characteristics	−0.0402***	43.36			0.0557***	76.32			0.0003	1.39		
	(0.0115)				(0.0077)				(0.0111)			
Unexplained: due to diff. in coefficients	−0.0525**	56.64			0.0173	23.68			0.0194	98.61		
	(0.0234)				(0.0146)				(0.0208)			
Total	−0.0927***				0.0730***				0.0197			
	(0.0205)				(0.0148)				(0.0176)			
Panel 2: Detailed decomposition												
	Explained		Unexplained		Explained		Unexplained		Explained		Unexplained	
Age	0.0014	−1.48	−0.0691	74.57	0.0036**	4.92	−0.0255	−34.96	0.0012	6.17	−0.0015	−7.41
	(0.0037)		(0.0702)		(0.0016)		(0.0337)		(0.0601)		(0.0370)	
Marital status	−0.0068	7.37	−0.0029	3.09	−0.0080***	−10.92	0.0149	20.41	−0.0047	−23.99	0.0520*	264.22
	(0.0048)		(0.0323)		(0.0021)		(0.0191)		(0.2342)		(0.0297)	
Primary education	0.0395***	−42.61	−0.1224***	132.00	−0.1191***	−163.09	−0.1442	−197.39	0.0051	25.82	0.0483	245.43
	(0.0134)		(0.0444)		(0.0098)		(0.1738)		(0.2516)		(0.0299)	
Secondary education	−0.0009	1.01	−0.0824**	88.88	0.0062***	8.54	−0.1171	−160.35	−0.0001	−0.32	0.0480*	243.88
	(0.0006)		(0.0388)		(0.0004)		(0.1392)		(0.0032)		(0.0271)	
College education	−0.0209**	22.54	−0.0389**	41.99	0.0970***	132.81	−0.0431	−59.01	0.0012	6.09	0.0223**	113.21
	(0.0101)		(0.0167)		(0.0078)		(0.0511)		(0.0590)		(0.0112)	
University education	−0.0170**	18.29	−0.0022	2.38	0.0669***	91.66	−0.0178	−24.40	0.0019	9.76	0.0029	14.62
	(0.0071)		(0.0074)		(0.0055)		(0.0211)		(0.0948)		(0.0041)	
Income 500 to 2499	−0.0196***	21.13	0.0597***	−64.44	0.0043***	5.95	0.0083	11.38	−0.0025	−12.87	−0.0235	−119.40
	(0.0035)		(0.0219)		(0.0016)		(0.0101)		(0.1255)		(0.0149)	
Income 2500 to 4499	−0.0008***	0.88	0.0111	−11.94	−0.0001	−0.20	0.0013	1.72	−0.0002	−1.03	−0.0043	−21.70
	(0.0003)		(0.0107)		(0.0001)		(0.0022)		(0.0100)		(0.0063)	
Income over 4500	0.0164**	−17.67	0.0146	−15.74	0.0055**	7.49	0.0010	1.36	0.0060	30.62	−0.0107	−54.19
	(0.0072)		(0.0123)		(0.0028)		(0.0022)		(0.2984)		(0.0095)	
Household size	0.0013	−1.41	−0.0176	19.02	0.0008*	1.16	0.0110	15.02	0.0004	2.26	0.0277	140.92
	(0.0010)		(0.0589)		(0.0005)		(0.0193)		(0.0220)		(0.0384)	
Farmer	−0.0025***	2.73	0.0384	−41.44	−0.0010***	−1.38	0.0018	2.44	−0.0005	−2.37	−0.0178	−90.31
	(0.0007)		(0.0252)		(0.0004)		(0.0053)		(0.0231)		(0.0143)	
Govt. employee	0.0003	−0.37	0.0073	−7.87	−0.0008	−1.10	0.0008	1.08	−0.0010	−5.30	−0.0033	−16.59
	(0.0056)		(0.0115)		(0.0020)		(0.0019)		(0.0514)		(0.0066)	
Business	−0.0183***	19.75	0.0378**	−40.73	0.0025	3.46	0.0020	2.68	−0.0031	−15.92	−0.0216*	−109.54
	(0.0061)		(0.0165)		(0.0025)		(0.0034)		(0.1556)		(0.0121)	
Wage labor	0.0005	−0.59	−0.0031	3.31	0.0006	0.83	−0.0002	−0.34	0.0003	1.69	0.0012	6.24
	(0.0015)		(0.0060)		(0.0007)		(0.0013)		(0.0165)		(0.0035)	
Services	−0.0013	1.41	0.0017	−1.82	0.0003	0.43	0.0004	0.55	−0.0002	−1.01	−0.0001	−0.48
	(0.0009)		(0.0035)		(0.0004)		(0.0008)		(0.0098)		(0.0020)	
Retired	0.0066***	−7.13	0.0043	−4.65	−0.0004	−0.57	0.0008	1.04	0.0016	7.98	−0.0011	−5.37
	(0.0025)		(0.0033)		(0.0009)		(0.0010)		(0.0776)		(0.0026)	
House is rented	−0.0019	2.03	−0.0583**	62.88	−0.0043***	−5.83	0.0013	1.75	−0.0025	−12.65	0.0336*	170.73
	(0.0035)		(0.0289)		(0.0016)		(0.0054)		(0.1234)		(0.0196)	
House is self-owned	0.0027	−2.94	−0.1111*	119.83	−0.0017**	−2.27	−0.0003	−0.47	−0.0001	−0.70	0.0502	255.25
	(0.0017)		(0.0653)		(0.0008)		(0.0125)		(0.0069)		(0.0390)	
Variable/Model	Charcoal	Electricity	Firewood									
Panel 2: Detailed decomposition												
	Explained		Unexplained		Explained		Unexplained		Explained		Unexplained	
Employer owns house	−0.0005	0.51	−0.0030	3.27	−0.0002	−0.25	−0.0021	−2.91	−0.0005	−2.56	−0.0007	−3.69
	(0.0055)		(0.0070)		(0.0023)		(0.0031)		(0.0249)		(0.0037)	
Positive perception	−0.0008	0.89	−0.0080	8.59	−0.0008	−1.09	−0.0063	−8.67	−0.0005	−2.32	−0.0111	−56.49
	(0.0020)		(0.0329)		(0.0008)		(0.0105)		(0.0227)		(0.0194)	

(*continued on next page*)

Table 7 (*continued*)

Variable/Model	Charcoal				Electricity				Firewood			
		Percent		Percent		Percent		Percent		Percent		Percent
Access to information	−0.0167***	17.97	−0.0052	5.57	0.0053**	7.26	−0.0017	−2.27	−0.0011	−5.56	−0.0041	−20.74
	(0.0062)		(0.0119)		(0.0027)		(0.0027)		(0.0546)		(0.0072)	
Women's energy initiatives	−0.0010	1.06	0.0138	−14.92	−0.0011**	−1.50	0.0032	4.40	−0.0005	−2.40	−0.0023	−11.92
	(0.0011)		(0.0088)		(0.0006)		(0.0041)		(0.0234)		(0.0044)	
Constant	0.0000		0.2830*	−305.20	0.0000		0.3291	450.60	0.0000		−0.1650	−838.06
	(0.0000)		(0.1693)		(0.0000)		(0.3926)		(0.0000)		(0.1012)	
Observations	1779		1779	1779	1779		1779	1779	1779		1779	1779

Notes: Standard errors in parentheses. No formal education is used as a reference group. Income less than ZMW500 is used as a reference group. ***, **, *Statistically significant at 1 %, 5 % and 10 % significance levels, respectively. The percent column gives the contribution of each variable in the model to the overall difference in respective energy fuel's use between male and female headed households.

income contexts: charcoal, electricity, firewood, and candles. By applying gender-disaggregated analyses and nonlinear Oaxaca-Blinder and Fairlie decomposition methods to nationally representative data, this study contributes rare, rigorous empirical evidence on gendered energy disparities in Sub-Saharan Africa. Specifically, it quantifies how much of the observed gender gaps in fuel adoption are attributable to socioeconomic endowments versus unexplained structural or behavioral factors.

Several key findings emerge. First, socioeconomic characteristics such as education, income, and housing tenure significantly shape fuel choices, but their effects differ systematically between male-headed households (MHHs) and female-headed households (FHHs). Second, FHHs rely more heavily on traditional fuels—particularly charcoal—-driven by resource constraints and large unexplained structural barriers. Third, MHHs enjoy a 7–8 percentage point advantage in electricity adoption, of which 75–89 % is explained by disparities in education and income. Although mean firewood use shows no statistically significant difference by household type, the unexplained component overwhelmingly dominates, pointing to gendered time burdens, social norms, and unmeasured constraints.

These findings make substantial contributions to feminist economics, energy transition theory, and just energy transition debates that extend well beyond Zambia. By demonstrating that 75–89 % of the electricity gap is explained by education and income while the charcoal and firewood gaps are predominantly unexplained, our results reveal how gender operates through multiple intersecting mechanisms. This advances feminist economics by moving beyond simply documenting disparities to decomposing the relative importance of resource constraints versus patriarchal norms, intrahousehold bargaining power, and women's limited control over household finances—even when women are formal household heads. The persistence of large unexplained components, especially for traditional fuels empirically validates feminist claims that women's constrained agency reflects not only economic disadvantage but also deeply embedded structural barriers. Such include norms that assign fuel collection and cooking to women, institutional biases in credit and technology access, and time poverty that hinders energy transitions even when they are economically feasible.

Our findings also complicate linear "energy ladder" models by showing that the pathway from traditional to modern fuels is neither uniform nor solely income driven. The dominance of unexplained factors in firewood use validates and extends fuel-stacking frameworks, illustrating how gender norms generate distinct transition pathways for women that persist even in better-resourced contexts. Through a capabilities lens, these results demonstrate how energy poverty directly constrains women's freedoms and well-being across health, education, and economic participation, thereby reproducing gender inequality. That educated women in low-electrification areas cannot translate their human capital into cleaner cooking and lighting fuels highlights how energy access itself is a foundational capability. Furthermore, our gendered decomposition approach offers rigorous quantitative support for intersectionality theory. It reveals how gender compounds with rural location, lower education, and insecure housing tenure to produce overlapping axes of marginalization.

These insights have clear relevance across the Global South, where similar patterns of time poverty, informal labor, traditional fuel dependence, and patriarchal norms remain widespread. By showing that large gender gaps in clean fuel adoption persist after controlling for observable characteristics, our evidence challenges technocratic energy transition approaches that prioritize infrastructure expansion while neglecting gendered norms and institutional barriers. The dominance of unexplained components demonstrates that achieving clean energy transitions requires addressing structural and normative constraints, not merely closing income or education gaps. This provides strong empirical backing for "just energy transition" frameworks, underscoring that SDG 7 (affordable and clean energy) and SDG 5 (gender equality) are inseparable and must be pursued together.

For Zambia's National Energy Policy, the findings suggest a package of integrated and gender-responsive interventions. To reduce the charcoal gap (NEP Objective 4), policymakers should combine subsidized clean cooking distribution with flexible, mobile-money-based payment plans targeted at FHHs and community-level social marketing campaigns to shift cultural norms around cooking responsibilities. Closing the explained portion of the electricity gap (NEP Objective 10) calls for scholarships and technical training for women alongside expanded rural electrification; women's energy cooperatives could pool resources and overcome information asymmetries. Gender-responsive tariff design—lifeline tariffs for FHHs, time-of-use pricing aligned with women's cooking schedules, and mobile-money financing for home solar systems would improve affordability and uptake. The large unexplained components highlight the need for deeper structural reforms: appliance financing tailored to FHHs' liquidity constraints. Such includes landlord–tenant regulations that allow renters to install solar systems, and electricity tariff and reliability reforms, especially urgent given Zambia's recent load-shedding crises of up to 21 h per day.

Collectively, achieving gender-equitable energy access in Zambia and elsewhere demands simultaneous action on resource deficits and entrenched structural barriers. The rigorous empirical decomposition and theoretical extensions we offered here provide a robust foundation for both policy design and future scholarly efforts toward more inclusive and just energy transitions across the Global South.

While this study provides useful insights into the energy access literature, an important limitation should be noted. The decomposition techniques employed nonlinear Oaxaca-Blinder and Fairlie decompositions capture associations rather than causal relationships. While the results highlight significant gender gaps in energy use, they should not be interpreted as evidence of direct causality. Potential endogeneity concerns remain, as the methods cannot fully account for possible interdependencies between explanatory variables and energy outcomes. Thus, the observed associations reflect correlations rather than causal effects. Nonetheless, by systematically quantifying the sources of gender disparities, this study offers actionable insights for inclusive energy policies in Zambia and comparable energy-poor settings in the Global South, emphasizing the need for multidimensional strategies that address both structural inequalities and behavioral dynamics.

CRediT authorship contribution statement

John N. Ng'ombe: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Cleopas G. Sambo:** Writing – review & editing, Visualization, Validation, Supervision, Resources, Project administration, Data curation, Conceptualization. **Fafanyo Asiseh:** Writing – review & editing, Visualization, Validation, Supervision, Investigation, Conceptualization. **Stephen Prah:** Writing – review & editing, Visualization, Validation, Methodology, Investigation, Data curation, Conceptualization. **Obed Quaicoe:** Writing – review & editing, Visualization, Validation, Conceptualization. **Wisdom Takumah:** Validation, Conceptualization.

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Data availability

Data is available on request from the corresponding author.

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